

JUDGING STANDARDS IN YEAR 10

MATHEMATICS

Assessment pointers validate teachers' professional judgement when reporting against a five-point scale. The pointers:

- are examples of evidence in relation to the achievement standard
- should be used with the annotated student work samples
- exemplify what *students may demonstrate* rather than a checklist of *everything they should do*.

YEAR 10 MATHEMATICS ACHIEVEMENT STANDARD

Number and Algebra

At Standard, students recognise the connection between simple and compound interest. They solve problems involving linear equations and inequalities. Students make the connections between algebraic and graphical representations of relations. They expand binomial expressions and factorise monic quadratic expressions. Students find unknown values after substitution into formulas. They perform the four operations with simple algebraic fractions. Students solve simple quadratic equations and pairs of simultaneous equations.

Measurement and Geometry

Students solve surface area and volume problems relating to composite solids. They recognise the relationships between parallel and perpendicular lines. Students apply deductive reasoning to proofs and numerical exercises involving plane shapes. They use triangle and angle properties to prove congruence and similarity. Students use trigonometry to calculate unknown angles in right-angled triangles.

Statistics and Probability

Students compare data sets by referring to the shapes of the various data displays. They describe bivariate data where the independent variable is time. Students describe statistical relationships between two continuous variables. They evaluate statistical reports. Students list outcomes for multi-step chance experiments and assign probabilities for these experiments. They calculate quartiles and inter-quartile ranges.

YEAR 10 MATHEMATICS ASSESSMENT POINTERS

	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Number and Algebra					
Number and Place Value	Extracts relevant information from word problems and applies this to interest calculations. Compares loans and investments and provides valid justification for the recommendations made. Solves problems involving multi-step inverse processes by extracting contextual information.	Extracts relevant information from text and tables to calculate interest for loans and investments where interest is not compounded annually. Compares the effect of changed conditions, including rate and the number of times that interest is compounded per year.	Applies formulas for simple and compound interest for integral years. Applies compound interest as repeated applications of simple interest.	Applies simple interest to all calculations.	Does not meet the requirements of a D grade.
Patterns and algebra	Uses an understanding of rule of order to perform mixed operations involving fractions with numerical denominators.	Simplifies sums and differences of simple algebraic fractions with numerical denominators, and including binomials in the numerator. For example: $\frac{x+2}{4} + \frac{x}{2} = \frac{3x+2}{4}$ Applies index laws to simplify products and quotients of algebraic terms including fractions with numerical denominators which arise from authentic situations. For example: $\frac{x^2}{2} \times \frac{x}{3} = \frac{x^3}{6}$	Simplifies sums and differences of simple algebraic fractions with numerical denominators and monomial numerators. For example: $\frac{x}{3} + \frac{x}{4} = \frac{4x+3x}{12} = \frac{7x}{12}$ or $\frac{x^3}{3} - \frac{x^2}{3} = \frac{x^3-x^2}{3}$ Simplifies simple products and quotients of algebraic terms including fractions with numerical denominators which arise from authentic situations. For example: $l^2 \times 2l = 2l^3 \text{ or } \frac{l}{2} \times \frac{l^2}{2} = \frac{l^3}{4}$	Simplifies sums and differences of simple algebraic fractions with like terms and a common numerical denominator. For example: $\frac{x}{5} + \frac{2x}{5} = \frac{3x}{5} \text{ or } \frac{2x^2}{3} - \frac{x^2}{3} = \frac{x^2}{3}$	Does not meet the requirements of a D grade.
	Expands and simplifies binomial expressions. For example: $(2x+3)^2 - 2(x+5)$	Expands and simplifies binomial products including those involving non-unitary coefficients or perfect squares. For example: $(2x-7)(3x+2) \text{ and } (x+4)^2$	Expands and simplifies simple binomial products with unitary coefficients. For example: $(x-3)(x+4)$	Uses the distributive property to expand some simple binomial expressions and practised binomial products. For example: $x(x+4), 3(x+4), (x+2)(x+3)$	

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Patterns and algebra	Factorises a range of monic trinomials including those which result from the removal of a common factor. For example: $3x^2 - 15x + 18 = 3(x^2 - 5x + 6)$ $= 3(x - 3)(x - 2)$ Factorises a difference of squares with a common factor. For example: $7x^2 - 7a^2 = 7(x + a)(x - a)$ Formulates and solves monic quadratic equations which arise from authentic situations.	Factorises a range of monic trinomials but sometimes with error. Factorises difference of squares, including where the numerical term is a perfect square. For example: $x^2 - 25 = x^2 - 5^2$ $= (x + 5)(x - 5)$ Identifies and applies specific values of a given formula from an authentic situation and evaluates the expression. Solves linear and simple monic quadratic equations which arise from substitution of values into formulas.	Factorises monic trinomials involving obvious binomial factors. For example: $x^2 + 7x + 12 = (x + 3)(x + 4)$ or $x^2 - 7x + 12 = (x - 3)(x - 4)$ Substitutes values into formulas and simplifies. Solves simple linear equations which arise from substitution of values into familiar formulas.	Factorises binomial expressions by removing a common factor. For example: $3x^2 - 6x = 3x(x - 2)$ Substitutes values into familiar formulas and simplifies. For example simple interest or compound interest formula.	Does not meet the requirements of a D grade.
Linear and non-linear	Formulates and solves a linear inequality from a word problem, verifies the solution and graphs the solution on a number line. Solves problems involving parallel and perpendicular lines.	Represents and solves linear equations and simple inequalities related to authentic situations, including those which arise from inverse use of formula. Graphs the solutions resulting from problems involving simple linear inequalities. Verifies linear functions as being parallel or perpendicular by referring to the gradients.	Solves simple problems involving authentic linear equations or inequalities which are given or arise from inverse use of formula. Identifies lines and linear graphs as being parallel or perpendicular by referring to placement on the Cartesian plane	Solves given linear equations involving two operations. Solves given linear inequality involving two operations and non-negative factors.	Does not meet the requirements of a D grade.

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Linear and non-linear		Applies translations and reflections to sketch parabolas, circles and exponentials and links the algebraic and graphical forms of these functions.	Uses the Cartesian plane to compare families of lines and parabolas to demonstrate the connection of vertical and horizontal translations (no dilations). Sketches graphs of simple relations of circles and exponentials, using technology to assist. Matches graphs and equations for simple linear, quadratic, circular and exponential relations.	Identifies the form of the equation which matches linear, quadratic, circular and exponential relations from their graphs.	Does not meet the requirements of a D grade.
	Formulates a pair of simultaneous linear equations from a word problem, including defining the variables, and chooses an appropriate method to determine the solution.	Formulates a pair of simultaneous linear equations from a simple word problem where the variables are defined, and determines the simultaneous solution using either algebraic or graphical as requested.	Uses algebraic and graphical techniques, including using technology, to solve a pair of simultaneous linear equations and associates the algebraic solution to the point of intersection of their corresponding graphs.	Determines the simultaneous solution when given a graph of a pair of linear relations.	
	Solves a variety of quadratic equations by choosing a suitable strategy, including factorisation or quadratic formula. For example: $3x^2 = 4, x(x - 4) = 4$ $x^2 - 8x - 4 = 0, (x - 2)^2 = 9$	Solves a variety of familiar monic quadratic equations algebraically, but sometimes with error.	Solves simple monic quadratic equations algebraically when in factored or readily factorised form, or graphically where the quadratic is not readily factorised.	Solves a quadratic equation graphically and when given in factored form. For example: $(x - a)(x - b) = 0$	

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Measurement and Geometry					
Using units of measurement	Uses information from an authentic situation to choose between surface area or volume to solve problems involving a range of composite solids	Calculates the surface area and volume of composite solids from authentic contexts.	Chooses the appropriate formula and calculates the surface area and volume of a simple composite solid, where a diagram is provided, by considering the individual prisms and cylinders from which it is made.	Chooses the appropriate formula to calculate the surface area and volume of a right prism or cylinder.	Does not meet the requirements of a D grade.
Geometric reasoning	Communicates the proof of a pair of similar or congruent triangles by setting out a sequence of logically connected statements and geometrical conventions. Justifies similarity of a pair of triangles in a composite diagram using the geometric properties of the whole diagram.	Uses the conditions of similarity or congruence and a sequence of statements to prove similar or congruent triangles in composite diagrams.	Uses ratios of sides and/or values of corresponding angles in a given diagram to prove similar or congruent triangles.	Identifies a pair of triangles as congruent or similar when the size of angles and lengths of sides are given.	Does not meet the requirements of a D grade.
	Draws a composite diagram from given information, including information such as congruent angles and sides, then uses congruency or similarity to prove conjectures regarding other properties of the composite figure. Adds a construction to an existing diagram to assist in a geometric proof or calculation.	Uses geometric properties and conditions of similarity or congruence to justify a sequence of steps to calculate unknown angles or lengths in a two-dimensional diagram.	Uses deductive reasoning to calculate missing angles or lengths in a two-dimensional diagram involving parallel/perpendicular lines and plane shapes. Uses familiar geometric properties for simple proofs involving a limited number of steps.	Draws obvious conclusions from information given in a geometric diagram involving parallel/perpendicular lines, triangles and quadrilaterals using a limited sequence of statements.	

	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Pythagoras and trigonometry	Draws a clear two-dimension diagram representing the information from a complex situation, involving direction or angles of elevation and depression, and applies trigonometric ratios to solve related problems.	Extracts information, from a situation involving right-angled triangles or interprets a given complex diagram, and applies trigonometric ratios to solve related problems.	Chooses and uses the correct trigonometric ratio from the information given in a diagram and calculates unknown angles and lengths of sides.	Uses given trigonometric ratios to calculate unknown angles in right-angled triangles.	Does not meet the requirements of a D grade.
Statistics and Probability					
Chance	Recognises whether events are dependent or independent and identifies the effect of this on the probability of compound and conditional events.	Chooses an appropriate method to list outcomes for a two or three-step chance experiment. Determines the probability of compound and conditional events by locating subsets within the sample space.	Lists outcomes for two and three-step chance experiments using tree diagrams, arrays and Venn diagrams as required, with or without replacement, and calculates the probability of outcomes.	Lists the total outcomes for familiar two-step chance experiments using tree diagrams or two-way tables. Calculates probabilities by counting individual outcomes.	Does not meet the requirements of a D grade.
Data representation and interpretation	Constructs accurate box plots of data from primary and secondary sources, identifying possible outliers and using technology appropriately.	Determines and uses the five number summary values to construct box plots of data from primary and secondary sources, using technology appropriately.	Demonstrates an understanding of the calculation of quartiles and interquartile range and the link to the related box plot, using technology appropriately.	Recognises medians are involved in the calculation of quartiles, and that the interquartile range is the distance between upper and lower quartiles.	Does not meet the requirements of a D grade.
	Compares data sets with reference to centre, spread, skewness, symmetry, quartiles and proportions of the data.	Compares parallel box plots with accurate reference to medians and interquartile range and suggests reasons, related to the context, for any differences.	Compares parallel box plots, with reference to some obvious features of the two graphs.	Makes obvious observations related to box plots.	
	Compares the relative merits of a box plot with its corresponding histogram or dot plot with reference to centre, spread, skewness and symmetry. Relates these observations to the context when interpreting a statistical report.	Compares the shape of box plots to corresponding histograms and dot plots based on centre, spread and extremes.	Makes obvious statements without demonstrating an understanding of the significance of statistical features when comparing the shape of boxplots to corresponding histograms and dot plots.	Makes some observations about individual graphs without comparison.	

	A	B	C	D	E
	Excellent achievement	High achievement	Satisfactory achievement	Limited achievement	Very low achievement
Data representation and interpretation	Suggests appropriate reasons for changes in the dependent variable over time with reference to relevant events.	Identifies and describes patterns/trends in bivariate numerical data where time is the independent variable.	Identifies and describes obvious trends in bivariate numerical data where time is the independent variable. Makes reference to this relationship in familiar contexts, e.g. the median house price has increased over the past 10 years.	Describes bivariate data over time as either increasing or decreasing.	Does not meet the requirements of a D grade.
	Constructs accurate scatter plots, informally describing the strength and direction of the relationship between two numerical variables, identifying possible outliers.	Constructs scatter plots for authentic data and draws conclusions about the relationship between two numerical variables.	Uses scatter plots to identify and describe obvious relationships between two numerical variables.		
	Analytically dissects reports within the context, intention and claim of the report explaining the pitfalls (or motivation) of misinterpreting results.	Uses statistical concepts to evaluate reports or claims from a variety of sources including media and recognises the pitfalls of misinterpreting results.	Evaluates statistical reports from a variety of sources including media by considering the claim in the light of the source of the data, the method of collection, the accuracy of the display and the statistical interpretation of the data.	Evaluates statistical reports from a variety of sources, including media, with scaffolding and/or explicit questioning.	

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